CASE STUDY 2: ATM TRANSACTION SEQUENCE SIMULATION

1. Problem Statement

An ATM allows users to check their account balance, deposit money, and withdraw money. The system should process transactions in sequence and update the balance after every valid transaction. It should also handle insufficient balance, invalid amounts, and invalid menu choices.

2. Objectives

- To develop an ATM simulation program using Python.
- To perform deposit and withdrawal operations.
- To display the current account balance.
- To handle invalid transactions and insufficient balance.
- To understand conditional statements and loops in Python.

3. Assumptions

1. The initial account balance is Rs. 5000.
2. The user can check the balance, deposit money, withdraw money, or exit.
3. Deposit and withdrawal amounts must be greater than zero.
4. A withdrawal is allowed only when sufficient balance is available.
5. The program continues until the user selects the Exit option.

4. Algorithm

1. Start the program.
2. Initialize the account balance to Rs. 5000.
3. Display the ATM menu.
4. Ask the user to enter a choice.
5. If the choice is 1, display the current balance.
6. If the choice is 2, accept the deposit amount and add it to the balance if it is valid.
7. If the choice is 3, accept the withdrawal amount and check whether sufficient balance is available.
8. If the withdrawal is valid, subtract the amount from the balance; otherwise, display an error message.
9. If the choice is 4, display a thank-you message and stop the program.
10. For any other choice, display an invalid-choice message.
11. Repeat the menu until the user exits.

12. Stop the program.

5. Flowchart

Start
↓
Initialize balance = Rs. 5000
↓
Display ATM Menu
↓
Enter choice
↓
Is choice 1?

- Yes → Display balance → Return to menu

Is choice 2?

- Yes → Enter deposit amount → Validate amount → Update balance → Return to menu

Is choice 3?

- Yes → Enter withdrawal amount → Check amount and available balance → Update balance or display error → Return to menu

Is choice 4?

- Yes → Display thank-you message → Stop

Otherwise → Display invalid choice → Return to menu

6. Pseudocode

```
START
   balance = 5000

   REPEAT
      DISPLAY ATM MENU
      INPUT choice

      IF choice = 1
         DISPLAY balance

      ELSE IF choice = 2
         INPUT deposit amount
         IF amount > 0
            balance = balance + amount
```

```
            DISPLAY updated balance
        ELSE
            DISPLAY invalid amount
        END IF

    ELSE IF choice = 3
        INPUT withdrawal amount
        IF amount <= 0
            DISPLAY invalid amount
        ELSE IF amount <= balance
            balance = balance - amount
            DISPLAY updated balance
        ELSE
            DISPLAY insufficient balance
        END IF

    ELSE IF choice = 4
        DISPLAY thank-you message
        EXIT LOOP

    ELSE
        DISPLAY invalid choice
    END IF

  UNTIL choice = 4
STOP
```

7. Python Source Code

```
balance = 5000

while True:
    print("\n--- ATM MENU ---")
    print("1. Check Balance")
    print("2. Deposit Money")
    print("3. Withdraw Money")
    print("4. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        print("Current Balance: Rs.", balance)

    elif choice == "2":
        amount = float(input("Enter deposit amount: "))

        if amount > 0:
            balance = balance + amount
```

```
            print("Deposit Successful")
            print("Updated Balance: Rs.", balance)
        else:
            print("Invalid Amount")

    elif choice == "3":
        amount = float(input("Enter withdrawal amount: "))

        if amount <= 0:
            print("Invalid Amount")
        elif amount <= balance:
            balance = balance - amount
            print("Withdrawal Successful")
            print("Updated Balance: Rs.", balance)
        else:
            print("Insufficient Balance")

    elif choice == "4":
        print("Thank you for using ATM")
        break

    else:
        print("Invalid Choice")
```

8. Test Cases and Outputs

Test Case 1: Successful Deposit

Input:

Enter your choice: 2
Enter deposit amount: 1000

Expected Output:

Deposit Successful
Updated Balance: Rs. 6000.0

Test Case 2: Successful Withdrawal

Initial Balance: Rs. 5000

Input:

Enter your choice: 3
Enter withdrawal amount: 500

Expected Output:

Withdrawal Successful
Updated Balance: Rs. 4500.0

Test Case 3: Insufficient Balance

Initial Balance: Rs. 5000

Input:

Enter your choice: 3
Enter withdrawal amount: 10000

Expected Output:

Insufficient Balance

Note: Run each test case separately with the initial balance set to Rs. 5000.

9 Output Screenshots

```python
balance = 5000

while True:
    print("\n--- ATM MENU ---")
    print("1. Check Balance")
    print("2. Deposit Money")
    print("3. Withdraw Money")
    print("4. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        print("Current Balance: Rs.", balance)

    elif choice == "2":
        amount = float(input("Enter deposit amount: "))

        if amount > 0:
            balance = balance + amount
            print("Deposit Successful")
            print("Updated Balance: Rs.", balance)
        else:
            print("Invalid Amount")

    elif choice == "3":
        amount = float(input("Enter withdrawal amount: "))

        if amount <= 0:
            print("Invalid Amount")
        elif amount <= balance:
            balance = balance - amount
            print("Withdrawal Successful")
            print("Updated Balance: Rs.", balance)
        else:
            print("Insufficient Balance")

    elif choice == "4":
        print("Thank you for using ATM")
        break

    else:
        print("Invalid Choice")
```

Outputs:

```
============== RESTART: C:/Users/sirit/OneDrive/Desktop/case2.py ==============

--- ATM MENU ---
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 2
Enter deposit amount: 1000
Deposit Successful
Updated Balance: Rs. 6000.0

--- ATM MENU ---
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice:
============== RESTART: C:/Users/sirit/OneDrive/Desktop/case2.py ==============

--- ATM MENU ---
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 3
Enter withdrawal amount: 500
Withdrawal Successful
Updated Balance: Rs. 4500.0

--- ATM MENU ---
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice:
============== RESTART: C:/Users/sirit/OneDrive/Desktop/case2.py ==============

--- ATM MENU ---
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
```

```
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice:
============== RESTART: C:/Users/sirit/OneDrive/Desktop/case2.py ==============

--- ATM MENU ---
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 3
Enter withdrawal amount: 10000
Insufficient Balance
```

10. Conclusion

Thus, the ATM Transaction Sequence Simulation was successfully implemented using Python. The program performs balance checking, deposits, and withdrawals. It also handles insufficient balance and invalid amounts. This case study demonstrates the use of loops, conditional statements, and arithmetic operations in Python.